Summer Program on Mathematical Modeling 2026

Modeling Spatial Population Growth Using Reaction-Diffusion PDEs

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Introduction

1. Introduction

Câu 1. (1,5 điểm) Xét miền phẳng $R = \{(x,y) \in \mathbb{R}^2 : 1 < x^2 + y^2 < 4\}$.
(a) Vẽ miền R. Hãy chỉ ra bằng hình vẽ R có biên chính quy.
(b) Chứng minh rằng $u(x,y) = \ln((x-2)^2 + (y-6)^2)$ là hàm điều hòa trong hình tròn $B = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 5\}$. Từ đó, tính tích phân thế vị $I = \iint_R u(x,y)dxdy$.

Câu 2. (4,5 điểm) Xét bài toán Cauchy cho phương trình truyền sóng

$$u_{tt}(x,y,z,t) = \Delta u(x,y,z,t) + \chi_{[0,1]}(t)\Big(\chi_{[2023,2024]}(x) - \chi_{[-2023,-2022]}(x)\Big), (x,y,z) \in \mathbb{R}^3, t > 0,$$

với các điều kiện ban đầu

$$u(x,y,z,0) = 0 \text{ và } u_t(x,y,z,0) = \chi_B(x,y,z),$$

trong đó $B = \{(x,y,z) \in \mathbb{R}^3 : x^2 + y^2 + (z-2026)^2 < 1\}$. Hãy tính $u(0,y,0,t), y \in \mathbb{R}, t > 0$.

Câu 3. (3,5 điểm) Xét bài toán biên cho phương trình Laplace:

$$u_{xx} + u_{yy} = 0 \text{ trong tam giác } T = \{(x,y) : 0 < y < 1 - |1-x|\},$$

với điều kiện biên $u(y,y) = y, u(2-y,y) = 0$ khi $0 < y < 1$, và $u_y(x,0) = 0$ khi $0 < x < 2$.
(a) Sử dụng phép phản xạ Schwarz qua trục hoành $y = 0$, bài toán đang xét chuyển thành bài toán nào trong hình vuông $Q = \{(x,y) : |y| < 1 - |1-x|\}$? Từ đó, chứng minh rằng $0 < u < 1$ trong T.
(b) Dùng "phép quay" $X = (x+y)/2, Y = (x-y)/2$, bài toán trong hình vuông Q ở ý (a) chuyển thành bài toán nào? Giải bài toán này, rồi quay lại giải bài toán ban đầu.

Câu 4. (2,5 điểm) Giải tường minh nghiệm bị chặn của bài toán giá trị ban đầu cho phương trình truyền nhiệt

$$u_t(x,y,t) = 2[u_{xx}(x,y,t) + u_{yy}(x,y,t)] + e^{-t}\cos(x)\sin^2(2y), x,y \in \mathbb{R}, t > 0,$$

điều kiện ban đầu $u(x,y,0) = 0, x,y \in \mathbb{R}$.

1. Introduction

Why this project?

Instead of studying ODEs and PDEs only from a theoretical perspective which is really boring and hard to understand, why not apply them to some interesting Mathematical-Biological Models?

Mathematical Modeling

2. Mathematical Modeling

When ODE and PDE combine

Exponential Population Growth Equation

Firstly, we start with a simple Exponential Population Growth Equation as below:

$$\frac{dN}{dt} = rN$$

Where $N(t)$ is the population size at time t, and r is the growth rate (can be a constant r or a function over time $r(t)$; to be simple we assume that $r = const$).

2. Mathematical Modeling

When ODE and PDE combine

Logistic Population Growth Equation

Now that with the Carrying Capacity K, we can use a Logistic Model:

$$\frac{dN}{dt} = r\left(1 - \frac{N}{K}\right)N$$

- $N \ll K$: The population grows exponentially.
- $N \to K$: The population's growth speed decreases.
- $N = K$: The population stops growing since $\frac{dN}{dt} = r\left(1 - \frac{N}{N}\right)N = 0.$

2. Mathematical Modeling

When ODE and PDE combine

Reaction-Diffusion Equation

Supposing that the population starts at the center of a fixed segment on the Ox axis and it gradually spreads. The population density at the point x at time t is described by $N = N(x,t)$:

$$\frac{\partial N}{\partial t} = \underbrace{D\frac{\partial^2 N}{\partial x^2}}_{\text{Diffusion}} + \underbrace{r\Big(1-\frac{N}{K}\Big)N}_{\text{Population Growth}}$$

By normalizing the population density as $u = N/K$, the equation becomes the *Fisher–KPP equation*.

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2} + ru\Big(1-u\Big)$$

Where $D > 0$ is the diffusion coefficient, which represents the rate at which the population spreads through space.

2. Mathematical Modeling

Expansion to Two-Dimensional Space

Reaction-Diffusion Equation on Two-Dimensional Space

Suppose that a population is initially concentrated at the center of a rectangular domain in the Oxy plane and gradually spreads in both the horizontal and vertical directions. The population density at position (x, y) and time t is described by $N = N(x, y, t)$:

$$\frac{\partial N}{\partial t} = D_x \frac{\partial^2 N}{\partial x^2} + D_y \frac{\partial^2 N}{\partial y^2} + r\left(1 - \frac{N}{K}\right)N$$

Where $D_x, D_y > 0$ are the diffusion coefficients representing the rates at which the population spreads in the horizontal and vertical directions, respectively.

2. Mathematical Modeling

Expansion to Two-Dimensional Space

Reaction-Diffusion Equation on Two-Dimensional Space

If $D_x = D_y$ then we can write in short as:

$$\frac{\partial N}{\partial t} = D\Big(\frac{\partial^2 N}{\partial x^2} + \frac{\partial^2 N}{\partial y^2}\Big) + r\Big(1 - \frac{N}{K}\Big)N$$

Or even shorter:

$$\frac{\partial N}{\partial t} = D\nabla^2 N + r\Big(1 - \frac{N}{K}\Big)N$$

Note that D_x, D_y, and D may be functions of space and time, such as $D_x(x, y, t)$, $D_y(x, y, t)$, and $D(x, y, t)$. However, for simplicity, we assume that they are constants.

2. Mathematical Model

Neumann Boundary Conditions

No-Flux Boundary Condition

We assume that the population cannot leave the domain. Therefore, there is no population flux across the boundary:

$$\frac{\partial N}{\partial n} = 0 \qquad \text{on } \partial\Omega$$

where $\partial\Omega$ denotes the boundary of the spatial domain.

For a Rectangular Domain

For $\Omega = [0, L_x] \times [0, L_y]$, the Neumann Boundary Condition is:

$$\frac{\partial N}{\partial x}(0, y, t) = \frac{\partial N}{\partial x}(L_x, y, t) = 0,$$

$$\frac{\partial N}{\partial y}(x, 0, t) = \frac{\partial N}{\partial y}(x, L_y, t) = 0.$$

2. Mathematical Model

Dirichlet Boundary Condition

Dirichlet Boundary Condition

Instead of specifying the population flux, we specify the population density directly on the boundary:

$$N(x, y, t) = g(x, y, t) \qquad \text{on } \partial\Omega.$$

For example, a zero-density boundary condition is:

$$N(x, y, t) = 0 \qquad \text{on } \partial\Omega.$$

Interpretation

The population density is prescribed at the boundary.
Unlike the Neumann condition, individuals may cross the boundary.
To be simple we use Neumann Boundary Condition in this presentation.

Numerical Methods

3. Numerical Methods

Spatial Discretization

Finite Difference Method

We discretize the spatial domain into a uniform grid:

$$x_i = i\Delta x, \qquad y_j = j\Delta y$$

and approximate the population density by

$$N(x_i, y_j, t) \approx N_{i,j}(t).$$

3. Numerical Methods

Second-Order Spatial Derivatives

Second-Order Spatial Derivatives

Using the central difference approximation

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

Then:

$$N_{xx} = \frac{\partial^2 N}{\partial x^2} \approx \frac{N_{i+1,j} - 2N_{i,j} + N_{i-1,j}}{(\Delta x)^2}$$

$$N_{yy} = \frac{\partial^2 N}{\partial y^2} \approx \frac{N_{i,j+1} - 2N_{i,j} + N_{i,j-1}}{(\Delta y)^2}.$$

3. Numerical Methods

Time Discretization

Let

$$t_n = n\Delta t, \qquad N_{i,j}^n \approx N(x_i, y_j, t_n).$$

Using the Explicit Euler method:

$$\frac{\partial N}{\partial t} \approx \frac{N_{i,j}^{n+1} - N_{i,j}^n}{\Delta t}.$$

Therefore,

$$N_{i,j}^{n+1} = N_{i,j}^n + \Delta t \Big[D_x N_{xx} + D_y N_{yy} + r N_{i,j}^n \Big(1 - \frac{N_{i,j}^n}{K}\Big)\Big],$$

Visualization and Simulation

4. Visualization and Simulation

One-Dimensional Simulation

Suppose we have a population of animals in a closed one-dimensional habitat

$$\Omega = [0, 100]$$

and no animal can cross the boundary of the domain. Therefore, the Neumann Boundary Condition is:

$$\frac{\partial N}{\partial x}(0, t) = \frac{\partial N}{\partial x}(100, t) = 0.$$

We choose the simulation constants:

$$D = 1, \qquad r = 0.1, \qquad K = 100.$$

4. Visualization and Simulation

One-Dimensional Simulation

We choose the initial population distribution:

$$N(x,0) = N_0 \exp\left[-\frac{(x-x_0)^2}{2\sigma^2}\right].$$

Suppose that the population initially concentrates at the center of the domain, where

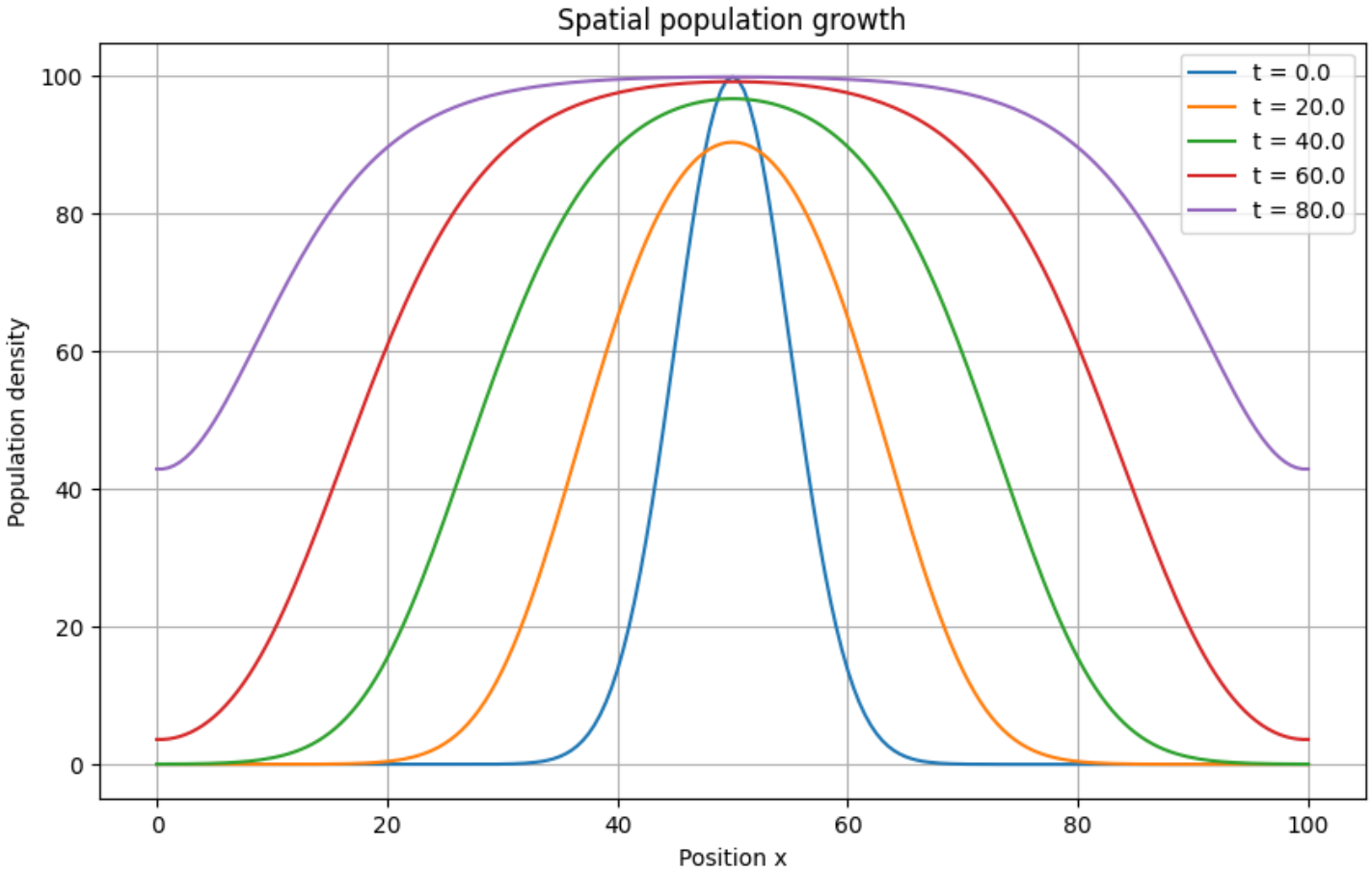
$$x_0 = 50, \qquad N_0 = 100, \qquad \sigma = 5.$$

Then the initial population distribution is:

$$N(x,0) = 100 \exp\left[-\frac{(x-50)^2}{50}\right].$$

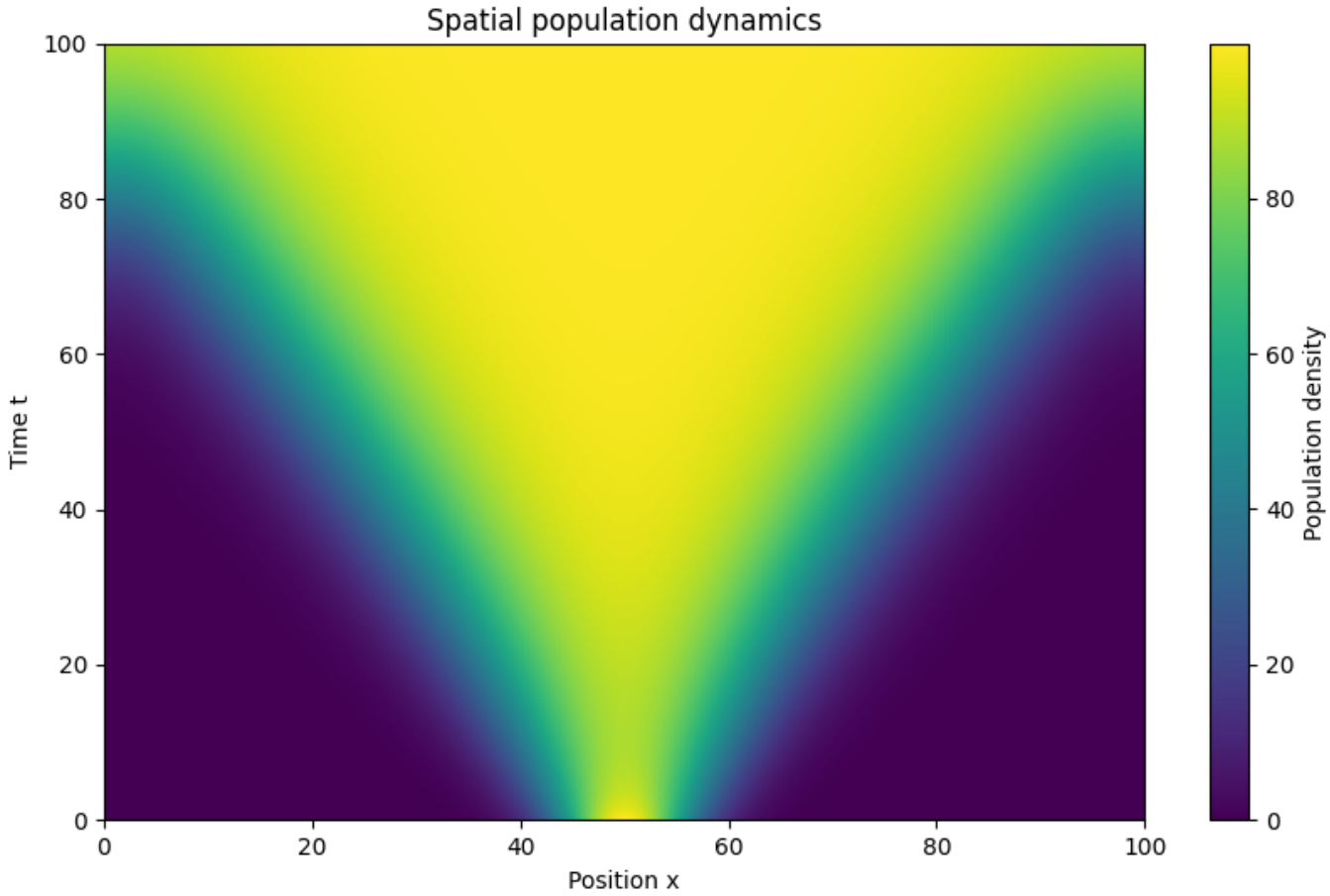
4. Visualization and Simulation

One-Dimensional Simulation



4. Visualization and Simulation

One-Dimensional Simulation



4. Visualization and Simulation

Two-Dimensional Simulation

Suppose we have a population of animals in a closed rectangular domain

$$\Omega = [0, 100] \times [0, 100]$$

and no animals can leave the domain. So the Neumann Boundary Condition is:

$$\frac{\partial N}{\partial x}(0, y, t) = \frac{\partial N}{\partial x}(100, y, t) = 0,$$
$$\frac{\partial N}{\partial y}(x, 0, t) = \frac{\partial N}{\partial y}(x, 100, t) = 0.$$

We choose simulation parameters:

$$D_x = 4, \qquad D_y = 1, \qquad r = 0.1, \qquad K = 100.$$

4. Visualization and Simulation

Two-Dimensional Simulation

And we choose the initial state:

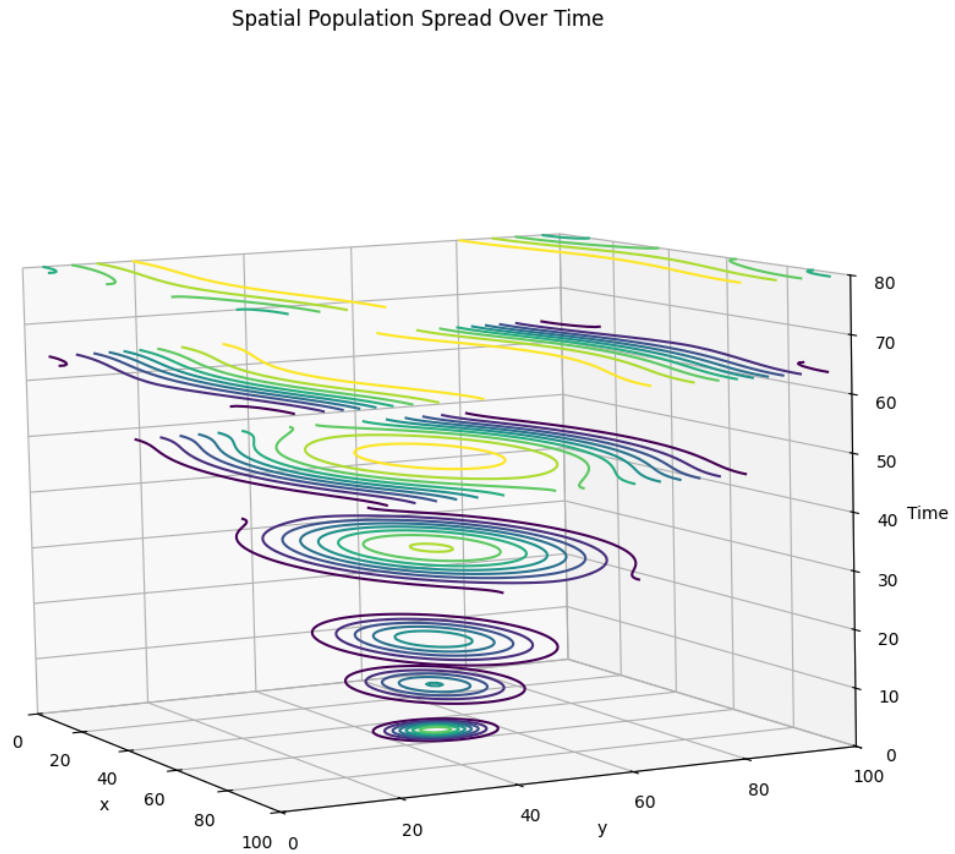
$$N(x, y, 0) = N_0 \exp\Big[- \frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2} \Big]$$

Suppose that the initial location is at the center of the rectangular domain $(x_0, y_0) = (50, 50)$, the maximum initial population density is $N_0 = 80$ and the initial population spread is $\sigma = 5$. Then we have:

$$N(x, y, 0) = 80 \exp\Big[- \frac{(x - 50)^2 + (y - 50)^2}{50} \Big]$$

4. Visualization and Simulation

Two-Dimensional Simulation



Conclusion

5. Conclusion

Conclusion

- We extended the Logistic Growth Model from an ODE to a Reaction-Diffusion PDE to describe spatial population dynamics.
- The model captures both population growth and spatial diffusion.
- Numerical simulations illustrate how the population spreads and approaches the carrying capacity over time.
- Different diffusion coefficients can produce different spatial spreading patterns.

References

References

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Thank You! Questions & Discussion

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